THE GEOMETRY OF UNCONFORMITIES (1)

*An **unconformity** is a surface of contact separating younger sedimentary rocks from older igneous, metamorphic or sedimentary rocks, along which there is evidence of a significant time-break (**hiatus**)* in the geologic record due to erosion and/or a break in the sedimentation.*

(* the short time-break represented by a bedding surface in a conformable succession is referred to as a *diastem*)

Unconformities are categorized on the basis of

(a) the shape of the unconformity surface and the orientation of the units on either side of it, or

(b) the orientation relationship between the units and the unconformity surface.

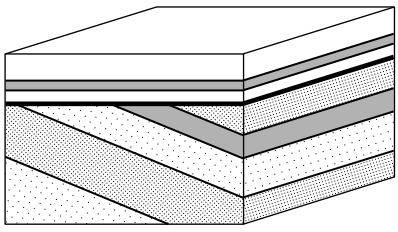
The former emphasizes the geometry of the unconformity whereas the latter is of significance in depositional sequence analysis

Descriptions emphasizing the geometry of the unconformity

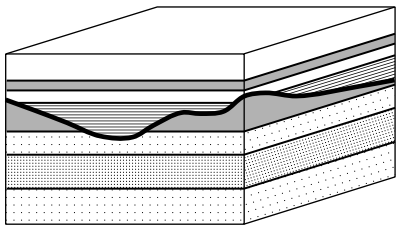
(1) An ***angular unconformity*** is one in which the younger sediments lie upon the eroded surface of tilted or folded older rocks. The implication is that the tilting or folding of the older rocks occurred in the time-break represented by the unconformity.

(2) A ***parallel unconformity*** is one in which the units on either side of the unconformity are in the same orientation, that is, have the same dip / dip direction. If the unconformity itself is a visible, irregular or uneven surface of erosion (such as a *buried landscape*), it is referred to as a ***disconformity***; if not and the unconformity is recognized only on the basis of a gap in the geologic record (*e.g.*, from missing biozones), it is referred to as a ***paraconformity***.

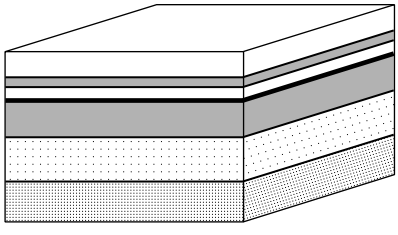
(3) A ***nonconformity*** is an unconformity in which the younger sediments lie on older igneous or metamorphic units which have been exposed to erosion in the time-break represented by the unconformity. A synonym is *heterolithic unconformity*.



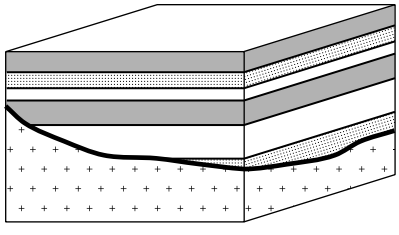
angular unconformity



disconformity



paraconformity



nonconformity

THE GEOMETRY OF UNCONFORMITIES (2)

Descriptions emphasizing the orientation relationship between the units and the unconformity surface

(A) Thinning of units against an unconformity surface is referred to as ***lapout***, and is described as

(A1) ***Toplap*** if the units below the unconformity thin against it (see below)

(A2) ***Baselap*** if the units above the unconformity thin against it. There are two types of baselap:

(A2a) ***Onlap*** (see below)

(A2b) ***Downlap*** (see below)

(B1) Descriptions emphasizing the orientation of the older units with respect to the unconformity surface:

(B1a) ***Erosional truncation*** where the unconformity is an erosional surface which cuts into the underlying units (*e.g.*, where the unconformity is the base of a channel)

(B1b) ***Toplap*** where the unconformity is marked by the truncation of initially inclined beds (*e.g.*, those formed on the foresets of deltas). Toplap typically results from non-deposition with minor erosion

(B1c) ***Concordance*** where the unconformity surface is parallel to the bedding surfaces in the older units

(B2) Descriptions emphasizing the orientation of the younger units with respect to the unconformity surface:

(B2a) ***Onlap*** where the younger units are approximately horizontal and the unconformity is inclined so that successively younger units rest on the unconformity surface. It arises when successively younger units are of greater lateral extent and thereby overlap the previously deposited units above the unconformity. Hence a synonym is *overlap*. As classically conceived, onlap occurs during relative rises in sea level (transgressions). (An *overstep unconformity* arises in similar manner when successively younger rocks rest upon progressively older members of the series under the unconformity, but as a description this has no depositional significance.)

(B2b) ***Downlap*** where the younger units are inclined and thin out down dip against the unconformity surface

(B2c) ***Concordance*** where the unconformity surface is parallel to the bedding surfaces in the younger units

(B3) Descriptions which refer to the retreat of the units above the unconformity surface away from it:

(B3a) ***Offlap*** where the units above the unconformity have progressively smaller lateral extents so that the upper units of the younger series extend less far than the units immediately below them

(B3b) ***Basinward shift*** which is a form of offlap but where the basinward shift in deposition occurs in discrete steps rather than progressively

Cross-section views

